Course Code : PHT 251 **MPNO/MS – 24 / 1568**

Fourth Semester B. Tech. (Electronics and Communication Engineering / Electronics Design Technology) Examination

INTRODUCTION TO ELECTROMAGNETIC THEORY

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory.
(2) Assume suitable data wherever necessary.
(3) Given Constants : $\epsilon_o = 8{\cdot}854 \times 10^{-12}$ F/m and $\mu_o = 4\pi \times 10^{-7}$ H/m.
(4) **Bold letters** represents vector.

1. Solve any **Six** questions :

(A) Find the total volume of the enclosing surfaces : $r = 2$ and 4, $\theta = 30^0$ and 50^0, $\phi = 20^0$ and 60^0. 2(CO1)

(B) Transform vector $2\mathbf{a}_x$ into cylindrical coordinate system at point :
$P(\rho = 4,\ \varphi = 120^0,\ z = 2)$. 2(CO1)

(C) A square plate residing in xy plane is situated in the space defined by $-3 \leq y \leq 3$. Find the total charge on the plate if the surface charge density $\rho_s = 4y^2\ \mu C/m^2$. 2(CO2)

(D) Elaborate Faraday's experiment. How the idea of flux density originate from it ? 2(CO2)

(E) What is Gauss law and Gaussian surface ? 2(CO2)

(F) Express Laplace operator in spherical coordinate system. 2(CO4)

(G) Explain magnetization and give importance of magnetic Permeability. 2(CO3)

(H) How polarization is significant in wave propagation ? 2(CO4)

2. (A) Flux density is given $D = \dfrac{Q\mathbf{a}_r}{4\pi R^2}$, find the charge enclosed by a closed surface. 3(CO2)

(B) Given the potential field, $V = 2x^2y - 5z$ and a point $P(-4, 3, 6)$. Find out the potential at point P, Electric field intensity E, Electric flux density D and volume charge density ρ_v. 4(CO3)

(C) Give the significance of electric field intensity in distributed form and write the expression for it. 3(CO2)

3. (A) Given three sheet current density :

   (i) $\mathbf{K_y} = 3\ \mathrm{nA/m^2}$ located at $x = 3$,

   (ii) $\mathbf{K_y} = -6\ \mu\mathrm{A/m^2}$ located at $x = -6$ and

   (iii) $\mathbf{K_y} = 10\ \mathrm{pA/m^2}$ located at $x = 15$, drawing current in y-direction.

   Find $\mathbf{H}$ at origin. 3(CO1,3)

(B) Given the field $D = 6\rho \sin 0{\cdot}5\varphi\ \mathbf{a_\rho} + 1{\cdot}5\rho \cos 0{\cdot}5\varphi\ \mathbf{a_\varphi}\ \mathrm{C/m^2}$, evaluate either side of the divergence theorem for the region bounded by $\rho = 2$, $\varphi = 0$, $\varphi = \pi$, $z = 0$ and $z = 5$. 4(CO1,2)

(C) Obtain and write down the Bio-Savart law in distributed form. 3(CO3)

4. (A) In a Material, for which $\sigma = 5\ \Omega^{-1}\mathrm{m}^{-1}$ and $\varepsilon_r = 1$, the electric field intensity is $E = 250 \sin[10^{10}(t)]$ V/m. Find the conduction and displacement current densities. Further find the frequency at which they have double magnitudes. 3(CO3)

(B) Consider a portion of sphere surface specified by $r = 4$, $0 \leqslant \theta \leqslant 0{\cdot}1\pi$ and $0 \leqslant \varphi \leqslant 0{\cdot}3\pi$. It is forming a closed path. The field is given by $\mathbf{H} = 6r \sin\varphi\ \mathbf{a_r} + 18r \sin\theta \cos\varphi\ \mathbf{a_\varphi}$.
Evaluate the either side of the stokes′ theorem. 4(CO2)

(C) State Ampere circuital law. Give its application and significance. 3(CO3)

5. (A) Given $E = E_m \sin(10^6 t - \beta z)\ \mathbf{a_y}$ in free space, find the expression for **D**, **B** and **H** at $t = 1\ \mu s$. 3(CO3)

(B) Obtain vector Helmholtz equation from phasor equation for electric field. 4(CO4)

(C) Ampere circuital law for static magnetic field has form Curl $H = J$. When it is extended to time varying field it changes its form, mention it and hence give the significance of Current displacement density. 3(CO4)

6. Write short notes on any **Two** :—

   (i) Maxwell's equations in time varying field.

   (ii) Point form and integral form.

   (iii) Polarization, linear, circular polarizations.

   (iv) Wave properties.

   (v) Vector magnetic potential and significance. 8(CO2,3,4)